

Innovation Points Report – MovieDB App

Project: Movie & Personnel Discovery App (TMDB-powered)

Module: Mobile Application Design

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Tech stack: Ionic 7.2.0 • Angular 17 (standalone) • TMDB API

In this report I've picked out the main innovation points from my app. I'm basically explaining what I built, how it works (at a high level), and why I did it that way. I focused on navigation, making the screens feel smoother to use, keeping performance decent, and giving it a clean dark UI.

1. Architecture & Navigation Logic

I noticed early on that I was repeating the same header stuff on multiple pages (title, back button, search, etc.), and it was getting messy fast. So I pulled it into one reusable component called `SharedHeaderComponent` and dropped it into the main screens. It keeps the layout consistent and also means I only have to fix/update the header in one place.

- **Header updates itself:** I listen for `NavigationEnd` and check the current route, so the header changes depending on where you are in the app. For example, when you go from the landing page into a person, the title/icon update automatically without me setting it manually on every screen.
- **One search box (no extra steps):** The search bar uses TMDB's multi search so users don't have to pick "movie" vs "person" first. I just check what type comes back and route it to the right page (movie details or person profile). It's a small thing, but it makes the app feel quicker.

2. User Experience (UX) Innovations

For UX I tried to avoid those moments where you click into something and then end up staring at a blank screen. A lot of my decisions here are about giving the user something useful to look at, even when there isn't a selected item loaded.

- **Fallback view instead of blank:** I used `*ngIf` so the Movie Details / Cast & Crew screens can switch between a normal details view and a History view. If the ID is null (like if you back out or refresh), it shows Recently Viewed items from `localStorage`. This stopped the "nothing here" problem I was getting during testing.
- **Filmography that actually reads in order:** TMDB credits sometimes come back in a weird order, so I added a `sortByYear` helper. I convert the date strings and sort newest-to-oldest, which makes it feel more like a timeline (and it's way easier to scan).
- **Read-more toggle for long bios:** Some biographies are massive, and they were pushing everything down the page. I added an expand/collapse "Read more" button with a simple height transition, so the default view stays neat but the full text is still there if you want it.

3. Performance & Optimization

Performance-wise, I kept things fairly lightweight by using Angular 17 standalone components and breaking features into smaller pieces. I also leaned on local storage for things like favourites/history so the app still feels responsive without needing extra calls.

- **Icons set up per page:** In Ionic 7 you have to register icons, so I call `addIcons` in the pages that use them. It kept my code a bit cleaner (and I wasn't accidentally importing loads of icons I never used).
- **Favourites stored locally (with duplicate checks):** I save favourites in `localStorage` so they don't disappear when the app closes. I also use `findIndex` to block duplicates because I kept adding the same movie twice when I was testing, and it made the list look broken.

4. Modern Visual Design

For the visual side, I went with a dark theme and a gold/charcoal palette because it suits movie posters and keeps the UI feeling modern. I tried to keep it stylish, but still readable (especially on smaller screens).

- **Glass-style info panels:** I made a "Glass Banner" style using `backdrop-filter: blur(15px)` and transparent layers. I used it on history/info sections so they stand out, but you can still see the background behind them.
- **Poster grid that adapts:** I used `ion-grid` with `size="4"` columns so the layout scales across devices. It helped keep posters lined up properly instead of jumping around when the screen size changes.